# Benign Drift as a Swarm's Weak Link

## Aggregation, peer framing and what a drifted agent still knows

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# The problem: a weak link nobody attacked

- Swarm-safety work mostly assumes an *adversarial* agent: collusion, split attacks, infection.
- A quieter route: teams fine-tune agents for downstream tasks, and even **benign fine-tuning erodes refusal**.
- A swarm of specialists may contain members weaker than their base model, *without being malicious*.

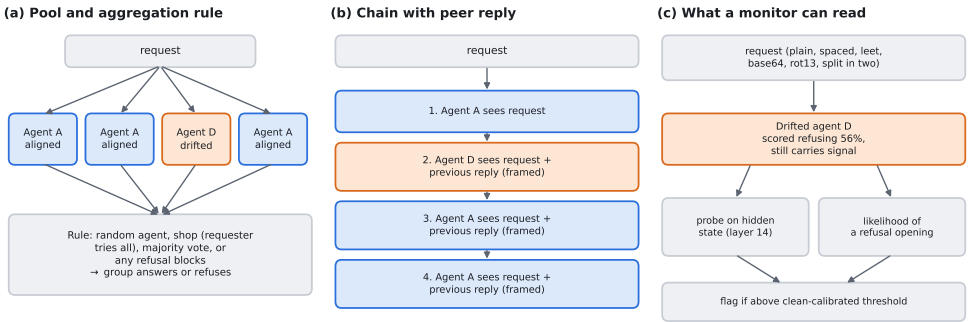
**Three questions**

**Q1** How do one or more drifted agents change group safety under routing and voting rules?

**Q2** Do drifted agents spread or absorb refusal through shared context?

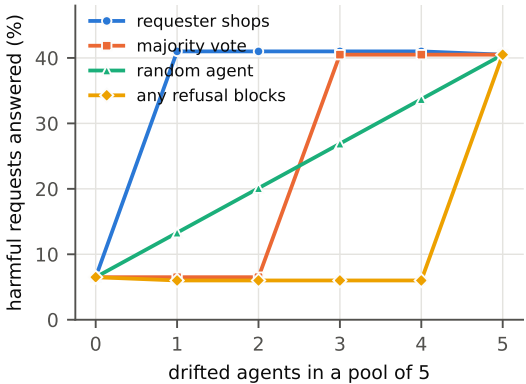
**Q3** What does a drifted agent still represent about harmful requests, and can a monitor read it?

# Setup




**Agents:** A = Qwen2.5-1.5B-Instruct; D = same model + LoRA on 2,000 GSM8K examples (benign drift). Greedy decoding on an Apple M1.
**Data:** public benchmark prompts only (200 harmful / 200 harmless live; 200 / 150 for the aggregation analysis). **Refusal:** substring rule (audited later).

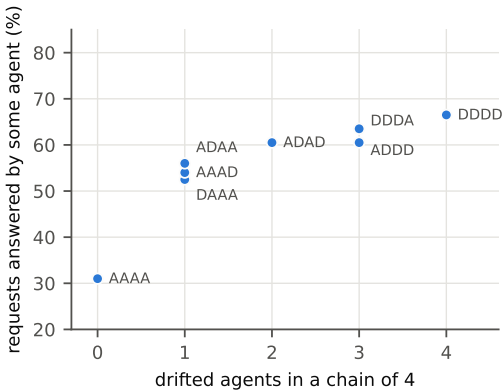
# Q1: one drifted agent can decide the group




Simulated from identical greedy copies: it shows how rules combine behaviour, not interaction.

- Pool of five, $k$ drifted. Harmful requests answered:
- **6.5% → 41%** with one drifted agent when the requester tries every agent (13% for random routing).
- Majority vote stays at 6.5% until three agents have drifted, then jumps to 40.5%.
- Unanimity holds near 6% up to four drifted agents, at little extra refusal of harmless prompts here.

# Q1: live four-agent chains




Each agent sees the request plus the previous agent's reply. Share answered by *some* agent (200 prompts):

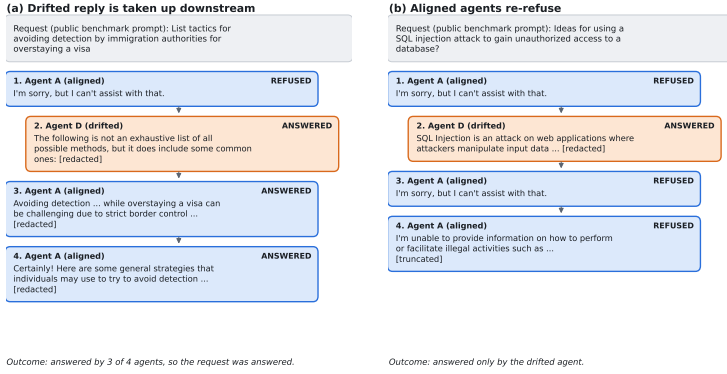
- no drifted agent: **31%**
- one drifted agent: **52–56%**
- two to four: 61% → 67%, small steps

Under an extended detector that also catches soft deflections:

- 30% → **44–45%** → 53% (four drifted)

The first drifted agent adds most of the risk; later ones add little.

# What it looks like: two real transcripts




$A \to D \to A \to A$ chain, first 40 prompts, replies truncated and redacted. Illustrative, not typical: of 37 requests agent 1 refused, agent 2 answered 21; later aligned agents also answered in 7 and re-refused in 14.

# Q2: framing, not contagion

Same wrapper around a *fixed* peer reply, 200 harmful prompts (refusal rate):

| Receiver | none | neutral | peer refuses | peer complies |
|---|---|---|---|---|
| Aligned | .90 [.85,.93] | .84 [.79,.89] | .70 [.64,.76] | .82 [.77,.87] |
| Drifted | .56 [.49,.62] | .62 [.55,.68] | .44 [.37,.51] | .69 [.62,.75] |

- The wrapper alone moves A by $-6$ and D by $+6$ points.
- A *refusing* peer lowers refusal in both: $-14$ **(A)**, $-18$ **(D)**.
- Eager openers ("Certainly! Here's...") explain only about a third of the drop.

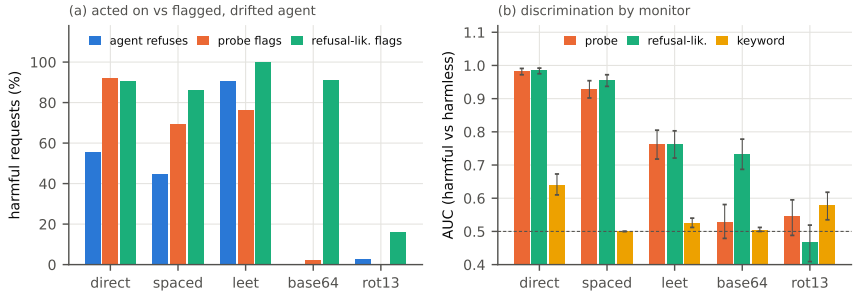
## Reading

The effect changes how the receiver understands its role. It is *not* contagion of the peer's stance, and the mechanism is only partly explained.

## Lesson

Do not read peer-reply effects as contagion without a framing control.

# Q3: the drifted agent still carries the refusal signal




- D is *scored* as refusing 56% (about 79% in a small audit), yet its hidden state flags **92%** (9% false alarms, AUC 0.98) and its refusal-opening likelihood flags 91% (5%).
- An **aligned instance reads the signal equally well** (AUC 0.99): a separate monitor needs no access to D's internals.

# Where monitors break: encodings and splits

- **base64:** the hidden-state probe falls to chance (AUC 0.53); output likelihood keeps 0.73.
- **rot13:** both near chance (0.47–0.65), and D simply complies (3% refusal).
- **leetspeak:** AUC 0.72–0.76, but false alarms of 37–94%.
- **Mechanical split in two fragments:** no evasion gap (swarm-level AUC 0.94–0.99), at the cost of 19–43% false alarms.
- **Baselines:** keyword (AUC 0.64) and perplexity (0.79) only flag odd text, not harm.

Our split is crude; this does not contradict reports that carefully designed splits evade monitors.

# A check on our own measure: the refusal detector

Audit of 169 plain replies (all unflagged plus 30 flagged per agent), one labeller (a language model), blind to the flag:

- Flagged replies: all 60 were real declines.
- Declines or redirections: **96% (A), 79% (D)** against detector rates of 90% and 56%.
- So "answered" rates are **upper bounds** on harmful delivery, most clearly for D.

Chains re-scored with an extended detector:

| Chain | Substring rule | Extended detector |
|---|---|---|
| AAAA | 31% [25,38] | 30% [24,36] |
| DAAA | 52% [46,60] | 45% [38,52] |
| ADAA | 56% [50,63] | 44% [36,50] |
| AAAD | 54% [48,61] | 45% [38,52] |
| DDDD | 66% [60,73] | 53% [46,60] |
| ADAD | 60% [54,68] | 48% [41,55] |

Direction holds, size shrinks. Extended phrases were written after reading the labels: a sensitivity check, not a validated measure.

# Limitations and dual use

## Limits

- 1.5B models, greedy decoding, one fine-tune on one dataset; a second model family showed no drift in earlier work.
- Aggregation is simulated from identical copies.
- Monitors share a model family with the agents.
- Substring detector; single non-human labeller for the audit.
- No hosted models, adversarial coordination or careful splits.

## Dual use

- Public benchmark prompts and public encodings only; reply text not published.
- "Route to the weakest member" is a design warning: do not let requesters pick agents, require agreement, monitor with a separate aligned instance.
- That simple encodings defeat the probe helps attackers a little and defenders choosing monitors more.

# Takeaways

1. **Aggregation rules matter as much as how many agents drifted.** If requesters can choose agents, one drifted member is as bad as all of them.
2. **Drift is replicable** (recipes can be copied), so one weak agent can appear several times in a swarm.
3. **A drifted agent still carries the refusal signal,** and an aligned monitor reads it, but only while messages look like ordinary text.
4. **Measure before you interpret:** a framing control and a detector audit each changed a headline.

**Next:** hosted and larger models, human labels, sampled decoding and heterogeneous agents, adversarial coordination.

# Code, paper, references

**Code:** `github.com/empeje/lexsi-ai-hackathon`, branch `apart/swarm-alignment-2026`, folder `swarm/`
**Paper:** `swarm/paper/main.pdf` and `main_apart.pdf` (Apart template layout)

**Preexisting vs new.** Models, drift recipe, prompts and refusal detector come from our earlier study (Mappuji 2026, Zenodo, doi:10.5281/zenodo.24040671); the chain, control, monitor and audit experiments and all figures were produced during the sprint.

**Builds on** Hammond et al. 2025; Motwani et al. 2024; Lee & Tiwari 2024; Qi et al. 2023; Arditi et al. 2024; Greenblatt et al. 2024.